

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}, \quad \mathbf{x} = [q_1, \dots, q_N, u_1, u_2, u_3]^\top, \quad \mathbf{M} = \text{diag}(1, \dots, 1, m_1, m_2, m_3)$$

$$\mathbf{K}_{qq} = \text{diag}(\omega_{mn}^2) + \sum_i k_i \vec{\phi}_i \vec{\phi}_i^\top, \quad \mathbf{K}_{qu_i} = -k_i \vec{\phi}_i, \quad \mathbf{K}_{u_i u_i} = k_i, \quad \vec{\phi}_i = [\phi_{mn}(x_i, y_i)]$$

$$\mathbf{C}: \text{the same with } c_i \text{ in place of } k_i \text{ and without the } \omega_{mn}^2 \text{ term;} \quad \mathbf{f} = [F_0 \phi_{mn}(x_d, y_d), 0, 0, 0]^\top$$